

Assignment 3: Portfolio Optimization, Stress Testing, and Systemic Risk

Due: Wednesday, July 22, 2026 @23:59

Objective: build optimal and robust portfolios from your three stocks, stress test them, and place them in a system-wide risk context.

Required tasks:

- 1) From your three stocks' mean returns and covariance, trace the mean-variance efficient frontier, mark and report the minimum-variance and the tangency (maximum-Sharpe) portfolios, and plot it. Also report the equal-risk-contribution (risk-parity) portfolio and each asset's risk contribution and compare all of these with the equal-weighted portfolio.
- 2) Start from the market-implied equilibrium returns, which you reverse-optimize from your stocks' market-cap weights (reuse the market caps from Assignment 2) and your covariance. Encode two views (one relative and one absolute) and how confident you are in each, derive the Black-Litterman weights, compare them with the plain mean-variance weights, and explain why anchoring to equilibrium gives steadier and more intuitive allocations.
- 3) Build a robust portfolio that allows for estimation error, for example a shrinkage-covariance minimum-variance portfolio or a resampled allocation. Explain why the tangency portfolio built from historical means swings so wildly with small changes in the inputs, and how risk parity and robust methods correct that instability.
- 4) Apply an ESG tilt or constraint to your portfolio and quantify its risk-return cost or benefit. Use real ESG or sustainability scores if you can source them, otherwise assign documented proxy scores and state where they came from. Where you can, take scores from two different providers or proxies and show how the disagreement between them changes which names your ESG portfolio favours.
- 5) Construct at least three plausible scenarios from macro-financial or ESG/climate risk factors, justified for your specific companies and sectors. Map each scenario to per-stock shocks, quantify the impact on portfolio value and on VaR and ES, compare the stress losses with your normal-times VaR, and add a historical stress window sliced from your own data. Judge whether your scenarios capture the kind of system-wide, correlated stress that macroprudential regulation is concerned with.
- 6) Compute a systemic-risk measure for one of your stocks against the market, for example a CoVaR or delta-CoVaR, or analyze how the correlations among your names spike in a crisis window (contagion), so that diversification fades exactly when it is most needed. Discuss the

model risk running through your whole exercise and where your models would be most likely to fail.

- 7) Run your scenarios on every candidate portfolio (equal-weight, minimum-variance, tangency, Black-Litterman, risk-parity, robust and ESG-constrained) and show which construction is most resilient. Then, using an out-of-sample split (estimate the weights on an earlier window and evaluate them on a later one), test whether any of your optimized portfolios actually beats the naive equal-weight ($1/N$) portfolio out of sample, as in DeMiguel, Garlappi and Uppal (2009). Make a final recommendation for your investor and assess the whole exercise against regulatory stress-testing frameworks (Basel III) and risk governance.

Deliverable: Jupyter section: optimization and scenario code, quantitative results, risk implications, commentary. $\leq 10,000$ chars.